

EXPERIMENT $\rightarrow$ OUTCOME

SET OF ALL OUTCOMES = SAMPLE SPACE

SET OF OUTCOMES $\Rightarrow$ EVENT

$$P(\text{EVENT}) = ?$$

IN A DISCRETE SAMPLE SPACE
WITH A FINITE # OF OUTCOMES

"FAIR" $\rightarrow$ EACH OUTCOME IS EQUALLY
LIKELY.

Roll a fair die, $P(\{1\}) = \frac{1}{6}$.

$$P(\text{EVENT}) = \frac{\# \text{ DESIRABLE OUTCOMES}}{\# \text{ TOTAL OUTCOMES}}$$

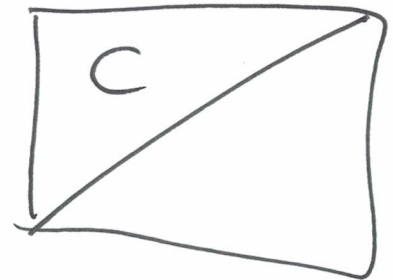
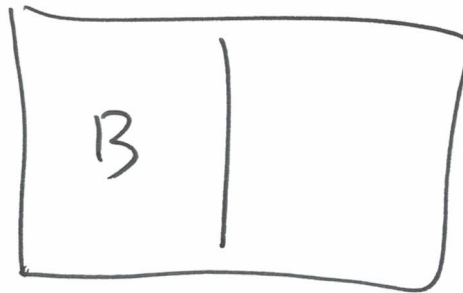
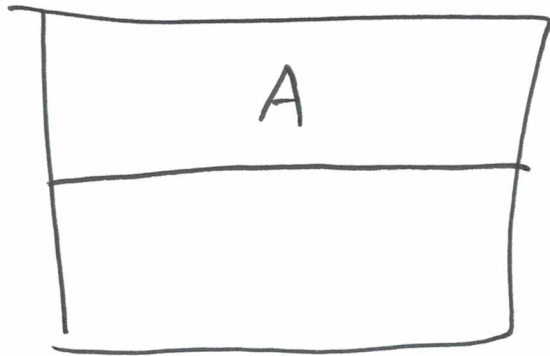
(*) ONLY
TRUE
FOR
A UNIFORM
DISTRIBUTION

$$P(\text{DIE IS EVEN}) = P(\{2, 4, 6\})$$
$$= \frac{3}{6} = \frac{1}{2}$$

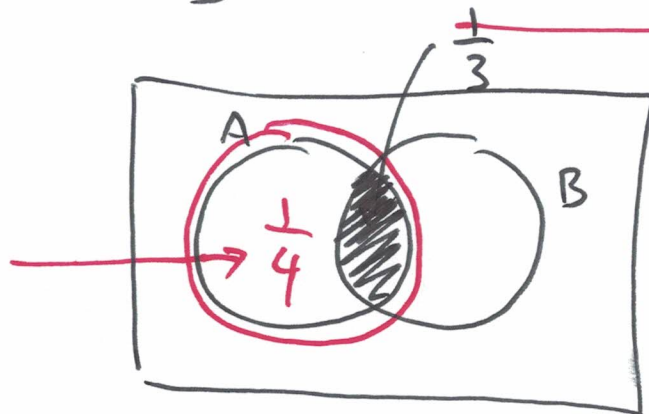
WE GENERALLY NEED TO ASSIGN
PROBABILITIES TO ALL THE EVENTS
THAT COULD OCCUR IN A WAY
THAT MAKES SENSE.

ARE THESE SITUATIONS LEGAL?

$$P(A) = \frac{1}{2}, \quad P(B) = \frac{1}{2}, \quad P(C) = \frac{1}{2}$$

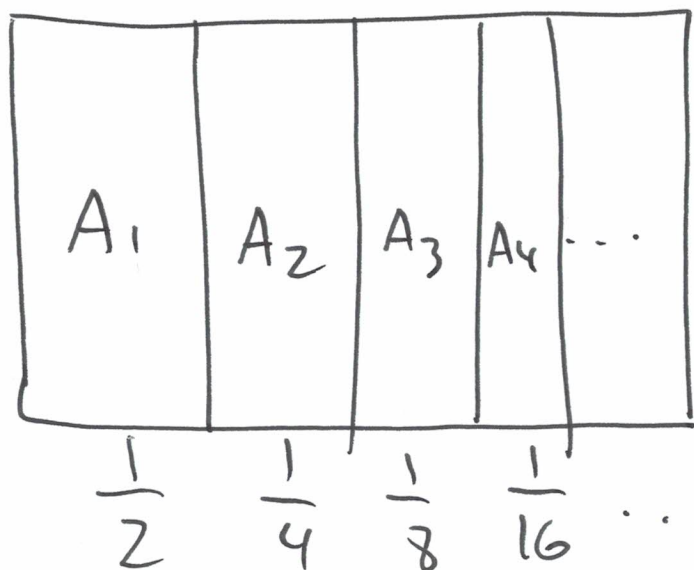


$$P(A \cap B) = \frac{1}{3} \quad P(A^c) = \frac{3}{4}$$



$$P(A) = \frac{1}{4}$$

X NOT POSSIBLE.



$$\bigcup_{i=1}^{\infty} A_i = S$$

$$A_i \cap A_j = \emptyset$$

A.K.A. A PARTITION OF S .

$$P(A_1) = \frac{1}{3}$$

$$P(A_2) = \frac{1}{9}$$

$$P(A_3) = \frac{1}{27}$$

$\vdots$

$$P(A_i) = \left(\frac{1}{3}\right)^i \quad \text{LEGAL?}$$

$$\sum P(A_i) =$$

$$\frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots$$

GEOMETRIC SERIES:

$$\frac{1}{3} + \frac{1}{9} + \frac{1}{27} + \dots$$

$$= \left(\frac{1}{3} \right) \left(1 + \frac{1}{3} + \frac{1}{9} + \dots \right)$$

$$\boxed{\sum_{k=0}^{\infty} a^k = \frac{1}{1-a}}$$

$$\rightarrow \frac{1}{1-\frac{1}{3}} = \frac{3}{2}$$

$$= \frac{1}{3} \cdot \frac{3}{2} = \frac{1}{2}$$

EX FLIP A COIN UNTIL YOU GET HEADS.

$$P(\text{HEAD}) = \frac{1}{3}$$

$X =$ # OF FLIPS.

$$P(X=1) = \frac{1}{3}$$

$$P(X=2) = \frac{2}{3} \cdot \frac{1}{3}$$

$$P(X=3) = \frac{2}{3} \cdot \frac{2}{3} \cdot \frac{1}{3}$$

$$P(X=k) = \left(\frac{2}{3}\right)^{k-1} \cdot \frac{1}{3}$$

A.K.A. GEOMETRIC DISTRIBUTION.

$$\begin{aligned} & \frac{1}{3} + \frac{2}{9} + \frac{4}{27} + \dots \\ & \frac{1}{3} \left(1 + \frac{2}{3} + \left(\frac{2}{3}\right)^2 + \dots \right) \\ & = \frac{1}{3} \cdot \frac{1}{1 - \frac{2}{3}} \\ & = \frac{1}{3} \cdot \frac{1}{\frac{1}{3}} = 1. \end{aligned}$$

$k = 1, 2, \dots, \infty$

PERMUTATIONS:

k SLOTS

n POSSIBILITIES PER SLOT

NO REPLACEMENT.

$$\begin{array}{cccccc} \text{---} & \text{---} & \text{---} & \text{---} & \text{---} & \text{---} \\ 6 & 5 & 4 & \cdot & 3 & \cdot & 2 & \cdot & 1 & = 720 \end{array}$$

IF n SLOTS = $n!$ POSSIBILITIES

$$\begin{array}{cccc} \text{---} & \text{---} & \text{---} & \text{---} \\ 6 & 5 & 4 & 3 \end{array}$$

$\frac{6!}{2!}$

$$\frac{n!}{(n-k)!}$$

e.g. 6 ITEMS
4 SLOTS

CHOOSE ~~k~~ A SET OF k OUT OF
 n POSSIBILITIES; ORDER DOESN'T MATTER.

$$\binom{n}{k} = \frac{n!}{(n-k)! k!}$$

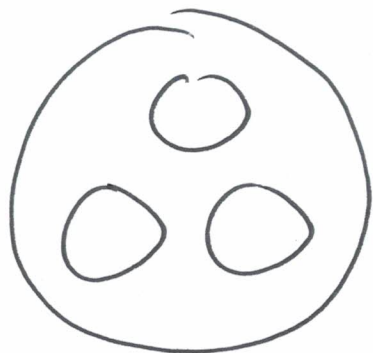
" n CHOOSE k "

MISSION REQUIRES 5-PERSON AWAY TEAM.

10 CANDIDATES. How MANY POSSIBLE TEAMS?

$$\binom{10}{5} = \frac{\overset{3}{10} \cdot \overset{2}{9} \cdot \cancel{8} \cdot 7 \cdot 6 \cdot \cancel{5} \cdot \cancel{4} \cdot \cancel{3} \cdot \cancel{2} \cdot \cancel{1}}{(\cancel{5} \cdot \cancel{4} \cdot \cancel{3} \cdot \cancel{2} \cdot \cancel{1})(\cancel{5} \cdot \cancel{4} \cdot \cancel{3} \cdot \cancel{2} \cdot \cancel{1})}$$

$$\frac{10!}{5!5!} = 3 \cdot 2 \cdot 7 \cdot 6 = \overset{42}{\underset{6}{\boxed{252}}}$$



PAD HAS 3 SLOTS

How MANY WAYS TO

TEAM THE TEAM DOWN?

A B C D E $\hookrightarrow$ $\binom{6}{3} = \frac{6 \cdot 5 \cdot 4}{\cancel{3 \cdot 2 \cdot 1}} = 20$

MONSTER EATS TEAM. HOW MANY
ORDERS OF EATING?

$$5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 = 120$$

SHIP FIRES ON MONSTER. AT LEAST
2 ~~SHOTS~~ HITS REQUIRED TO KILL.

SHIP FIRES 3 SHOTS TOTAL.

O = MISS

X = HIT

$$P(\text{HIT}) = 0.85$$

$$P(\text{MONSTER DIES}) = \frac{\cancel{O O O} \rightarrow (0.85)^3}{X X X}$$

$$= P(3^{\text{EXACTLY}} \text{ HITS}) + P(\text{EXACTLY } 2 \text{ HITS})$$

$$P(\text{AT LEAST } k \text{ SUCCESSES OUT OF } n)$$

$$= P(\text{EXACTLY } k \text{ SUCCESSES})$$

$$+ P(\text{EXACTLY } k+1 \text{ SUCCESSES})$$

$$+ \dots P(\text{EXACTLY } n \text{ SUCCESSES})$$

$$P(\text{EXACTLY } 3 \text{ HITS}) = (0.85)^3$$

$$P(\text{EXACTLY } 2 \text{ HITS}) = \begin{array}{ccc} \times & \times & 0 \end{array}$$

$$\binom{3}{1} \cdot (0.85)^2 (0.15) \begin{array}{ccc} \times & 0 & \times \\ 0 & \times & \times \end{array}$$

10 PEOPLE TO TRANSPORT

CAR THAT ONLY HOLDS 4 PEOPLE.

HOW MANY BE POSSIBLE SEQUENCES OF
TRIPS ARE THERE (ORDER INSIDE CAR
DOESN'T MATTER).

4 4 2

4 3 3

$$\begin{pmatrix} 3 \\ 1 \end{pmatrix} \cdot \begin{pmatrix} 10 \\ 4 \end{pmatrix} \begin{pmatrix} 6 \\ 4 \end{pmatrix} + \begin{pmatrix} 3 \\ 1 \end{pmatrix} \begin{pmatrix} 10 \\ 4 \end{pmatrix} \begin{pmatrix} 6 \\ 3 \end{pmatrix}$$

↑
WHERE IS
2-PERSON
SET

↑
4 FOR
FIRST CAR

↑
4 FOR
SECOND CAR.

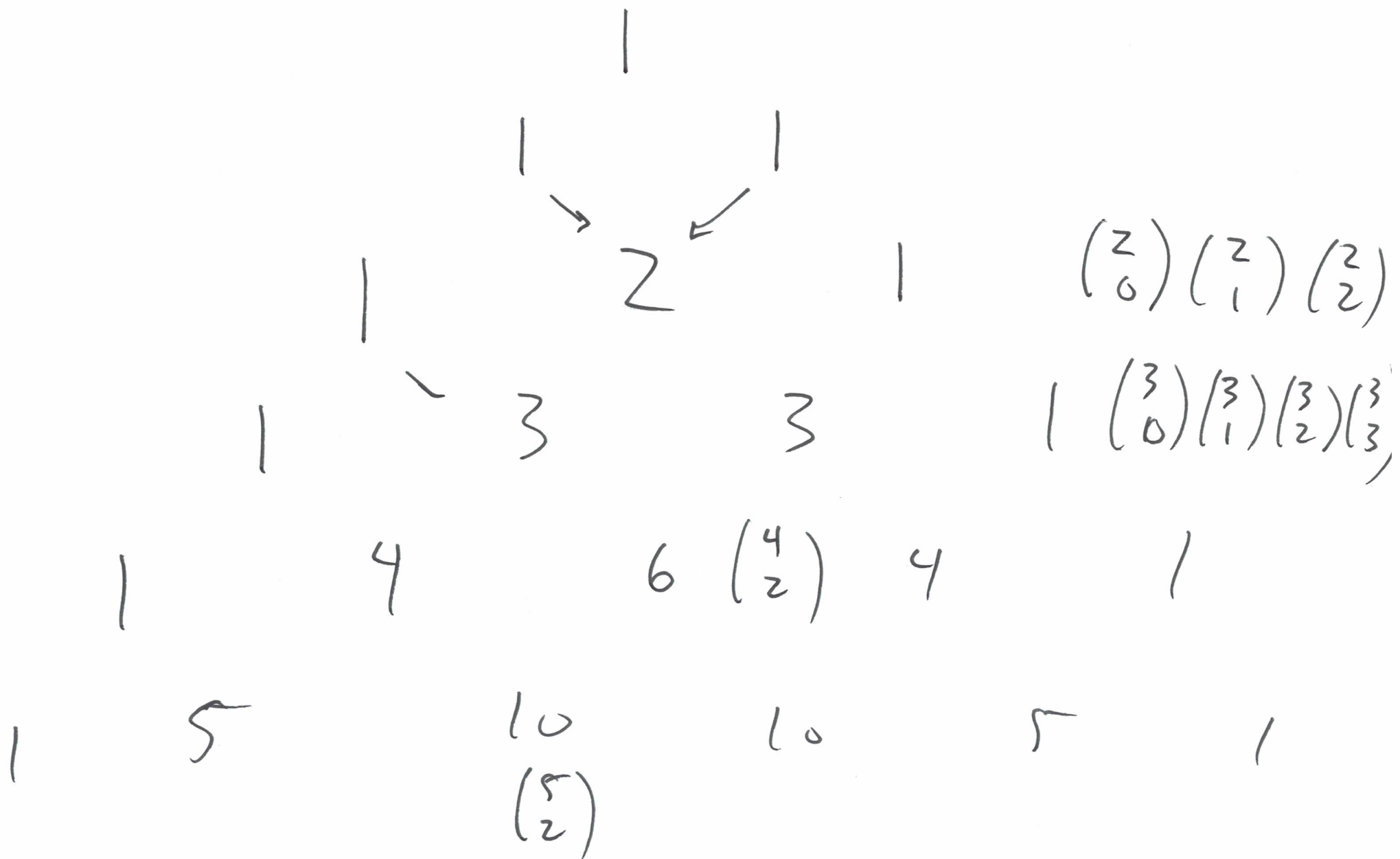
$$15 \quad \binom{10}{4} \binom{6}{3} = \binom{10}{3} \binom{7}{3} ?$$

$$\frac{10 \cdot 9 \cdot 8 \cdot 7}{\cancel{4} \cdot 3 \cdot 2 \cdot 1} \cdot \frac{6 \cdot 5 \cdot \cancel{4}}{3 \cdot 2} = \frac{10 \cdot 9 \cdot 8}{3 \cdot 2} \cdot \frac{7 \cdot 6 \cdot 5 \cdot \cancel{4}}{3 \cdot 2}$$

~~to 8~~

$$\binom{7}{3} = \frac{7 \cdot 6 \cdot 5 \cdot \cancel{4} \cdot \cancel{3} \cdot \cancel{2} \cdot 1}{(\cancel{4} \cdot \cancel{3} \cdot \cancel{2} \cdot 1) (3 \cdot 2 \cdot 1)}$$

$$\binom{7}{3} = \binom{7}{4}$$



3 2 3 2

$$\begin{pmatrix} 4 \\ 2 \end{pmatrix} \begin{pmatrix} 10 \\ 3 \end{pmatrix} \begin{pmatrix} 7 \\ 2 \end{pmatrix} \begin{pmatrix} 5 \\ 3 \end{pmatrix}$$

IF

SEATS

4
↓
IN

4.3.2

CAN

↑

ARE

NUMBERED.

4.3
—
2

SETUP: 6 NAMED BARBIES

PARTY STARTS WHEN ~~AT 4 NAMED~~^{ALL} BARBIES
ARRIVE. AT THIS POINT, NO MORE
KENS ARE ADMITTED. KENS ARE
INTERCHANGABLE.

$A = \{ \text{THERE ARE AT MOST } 0, 1, 2, \text{ or } 3 \text{ KENS AT FINAL PARTY.} \}$

HOW MANY OUTCOMES IN A

RECORD PARTY LIKE

$\{ B_1, K, K, K, B_2, B_3, K, B_4, B_5, B_6 \}$

0 KENS: 6 5 4 3 2 1 720

1 KEN 6 5 4 3 2 B
└────────────────────────────────┘
 $\binom{6}{1} \cdot 720$

2 KENS K K B
 $\binom{7}{2} \cdot 720$

3 KENS: $\binom{8}{3} \cdot 720.$

1

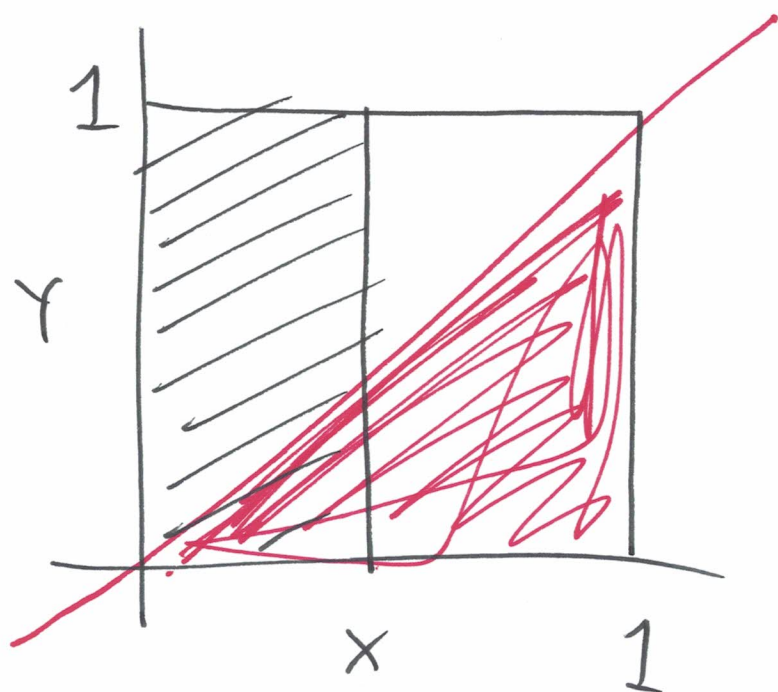
B_1 B_2 $[K_1, K_2]$ B_3 $[K_3]$ B_4

4 3 2 1

~~3 2 1~~

$$\binom{7}{3} \cdot 4! \cdot \boxed{\begin{matrix} 3 & 1 \\ & \cdot \end{matrix}}$$

WHAT IS EQUIVALENT OF "FAIR" IN
CONTINUOUS SAMPLE SPACE?

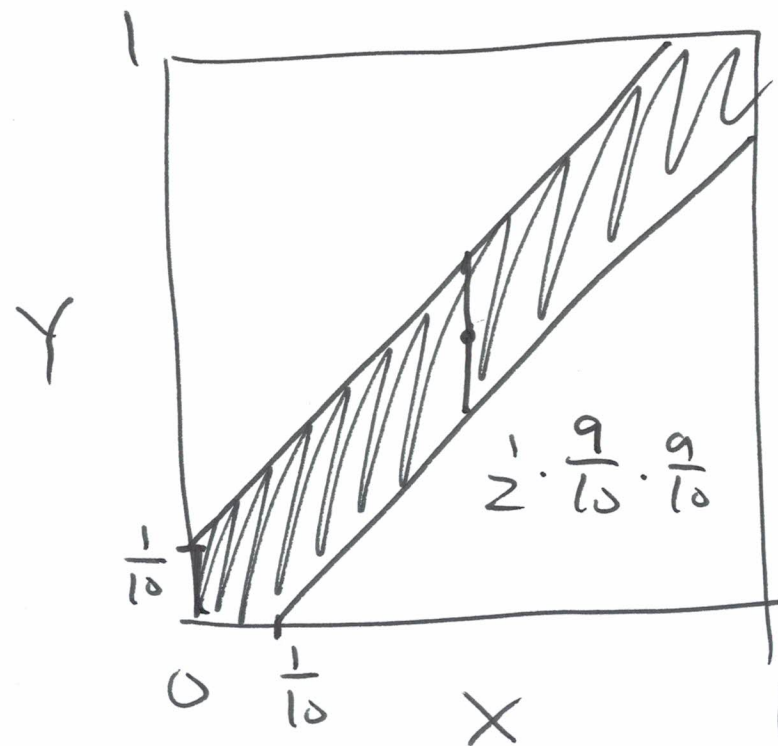


$$P\left(X < \frac{1}{2}\right) = \frac{1}{2}$$

$$P(X > Y) = \frac{1}{2}$$

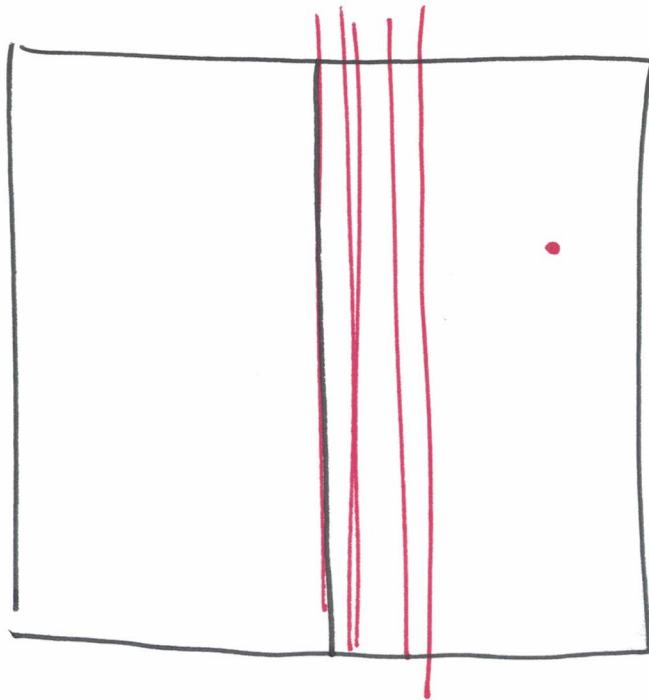
$$P(|X - Y| \leq \frac{1}{10}) =$$

UNIFORM DISTRIBUTION OVER THE SQUARE.



$$= 1 - \left(\frac{9}{10}\right)^2 = \frac{19}{100}.$$

IS $P(X \leq \frac{1}{2}) = P(X < \frac{1}{2})$ IN
CONTINUOUS SAMPLE SPACE?



$P(\text{AT LEAST 2 PEOPLE IN 100 PERIOD}$
 $\text{CLASS HAVE SAME BIRTHDAY})?$

$1 - \text{NO 2 PEOPLE HAVE SAME BIRTHDAY}$

$$\frac{365}{365} \cdot \frac{364}{365} \dots$$

$$\frac{365 - 100}{365}$$